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A RELATED WORK

Due to space constraints, we summarize related 3D and an extended list of 4D works in the appendix.

3D generation. Limited to single category trainings, early work on 3D generation extends GANs into 3D using a neural renderer as an inductive bias (DeVries et al., 2021; Chan et al., 2022; Or-EI et al., 2022; Schwarz et al., 2022; Bahmani et al., 2023b), Towards more diverse and flexible generation, CLIP-based supervision (Radford et al., 2021) enabled text-based generation and editing of 3D assets (Chen et al., 2018; Jain et al., 2022; Sanghi et al., 2022; Jetchev, 2021; Gao et al., 2023; Wang et al., 2022). With recent advances in diffusion models, Score Distillation Sampling (SDS) replaces CLIP supervision with diffusion model supervision (Poole et al., 2023; Wang et al., 2023d; Lin et al., 2023a; Chen et al., 2023c; Liang et al., 2023; Wang et al., 2023a; Li et al., 2024d; He et al., 2024b; Ye et al., 2024; Liu et al., 2024c; Yu et al., 2023b; Katzir et al., 2024; Lee et al., 2024a; Sun et al., 2024a) for higher quality generation. In order to improve the 3D structure of scenes, another line of work generates multiple views of a scene (Lin et al., 2023b; Liu et al., 2023; Shi et al., 2024; Feng et al., 2024a; Liu et al., 2024b; Kim et al., 2023; Voleti et al., 2024b; Höllein et al., 2024). Alternatively, other methods use iterative inpainting for 3D scene generation (Höllein et al., 2023; Shriram et al., 2024). Recent methods lift input images into 3D (Chan et al., 2023; Tang et al., 2023; Gu et al., 2023; Liu et al., 2024d; Yoo et al., 2023; Tewari et al., 2023; Qian et al., 2024c; Long et al., 2024; Wan et al., 2024; Szymanowicz et al., 2023) using NeRF (Mildenhall et al., 2020), 3D Gaussian Splatting (Kerbl et al., 2023), or Meshes in combination with diffusion models. Other recent methods (Hong et al., 2024; Li et al., 2024b; Xu et al., 2024d;c; Zhang et al., 2024a; Han et al., 2024; Jiang & Wang, 2024; Xie et al., 2024; Tang et al., 2024; Tochilkin et al., 2024; Qian et al., 2024a; Szymanowicz et al., 2024b;a) tackle fast feed-forward 3D generation, directly predicting a 3D generation from input images or text. Different from our approach, these methods can only synthesize static scenes.

4D generation. Previous methods also tackle the problem of 4D generation, i.e., generating videos of dynamic 3D scenes from controllable viewpoints, usually from an input text prompt or image. Since the initial work on this topic using large-scale generative models (Singer et al., 2023b), significant improvements in the visual quality and motion quality of generated scenes have been achieved (Ren et al., 2023; Ling et al., 2024a; Bahmani et al., 2024b; Zheng et al., 2024a; Gao et al., 2024b; Yang et al., 2024a; Jiang et al., 2024; Bahmani et al., 2024a; Zhang et al., 2024d; Xu et al., 2024a; Miao et al., 2024; Li et al., 2024a). While these methods generate scenes based on text conditioning, other approaches convert an input image or video to a dynamic 3D scene (Ren et al., 2023; Zhao et al., 2023; Yin et al., 2023; Pan et al., 2024; Zheng et al., 2024a; Ling et al., 2024a; Gao et al., 2024b; Zeng et al., 2024; Chu et al., 2024; Wu et al., 2024; Yang et al., 2024c; Wang et al., 2024; Feng et al., 2024b; Sun et al., 2024b; Zhang et al., 2024b; Yu et al., 2024; Ren et al., 2024; Lee et al., 2024b; Li et al., 2024c; Van Hoorick et al., 2024; Uzolas et al., 2024; Huang et al., 2024; Chai et al., 2024; Liang et al., 2024; Zhang et al., 2024c). Another line of work (Bahmani et al., 2023a; Xu et al., 2023) extends 3D GANs into 4D by training on 2D videos, however the quality is limited and models are trained on single category datasets. Still, all of these methods are focused on object-centric generation, typically based on 3D volumetric representations. As such, they typically do not incorporate background elements, and overall, they do not approach the level of photorealism demonstrated by the state-of-the-art video generation models used in our technique.

B QUANTITATIVE EVALUATION

We further evaluate all models for the task of single image-to-multiview generation. Furthermore, we provide results for established 2D video generation metrics.

B.1 MULTI-VIEW GENERATION

We evaluate our model for image-to-multiview generation. Due to ground truth correspondences for the RealEstate10K (Zhou et al., 2018) dataset, we can condition our model on a given camera trajectory and assess image based metrics, i.e., PSNR, SSIM, and LPIPS. We provide results for the low-resolution base model and the upsampled high-resolution results in Tab. 3.